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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Mid-Term Examination September – 2025

B.Tech, CSE/AI/CPS/DS/CS, 3rd Semester

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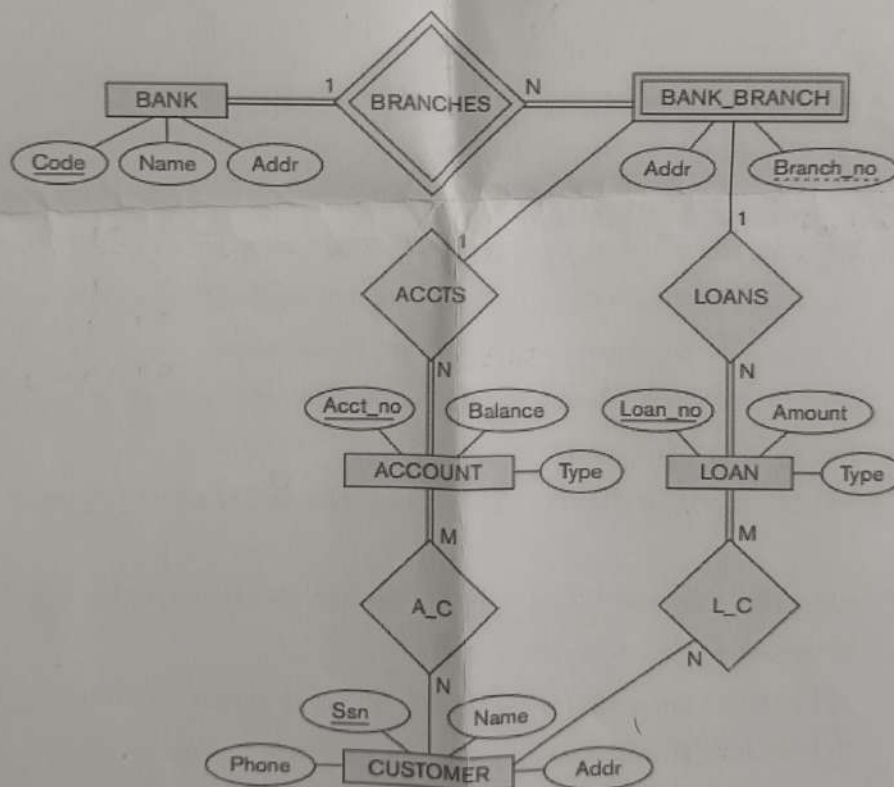
Subject Name/ Code: CSE/AI/CPS/DS/CS - 3002

MM:20

Time Allotted: 90 min

NOTE: 1. All questions are compulsory. 2. Mark answer number clearly.

- Q.1 (A) I. List out the difference between logical data independence and physical data independence? Which one is harder to achieve? (2)
- II. Demonstrate the usability of two tier and Three Tier architecture of Database Management System.



- Q.1 (B) Consider the ER diagram shown in above figure for part of a BANK database. Each bank (3)
- can have multiple branches, and each branch can have multiple accounts and loans.
- a. Identify all strong (non-weak) entity, their key attribute, in the ER diagram.
- b. Is there a weak entity type? If so, state its name, partial key, and identifying relationship.
- c. List all relationship types along with the participation constraints for each entity in the relationship. Justify your answers.

Q.2 (A) Consider the ER diagram provided in the figure above. Convert all entities and (3)
relationships into corresponding relational schemas. (Separate schemas for each
relationship and entity)

Q.2 (B) Using the relational schemas derived in Q.2 (A), merge them to reduce redundancy. (2)
Present the refined schemas along with all applicable constraints (e.g., keys, dependencies,
and referential integrity constraints). Ensure that all possible constraints (such as primary
keys, foreign keys, and participation constraints) are accurately modeled in the relational
schemas.

Q.3 (A) 1) Demonstrate the usage of divide operation in relational algebra. (2)
2) Why is union compatibility a fundamental requirement for performing set operations in
relational algebra?

Refer to the following relational schema for the subsequent questions.

Student(StudentID, Name, Branch, Year)

Project(ProjectID, Title, Domain, Duration)

Faculty(FacultyID, Name, Department, Designation)

Works_On(StudentID, ProjectID, Role)

Guides(FacultyID, ProjectID, Year)

Q.3 (B) Consider the schema provided in the above section to answer the below questions. (3)

- 1) Display the view of relations, illustrating all constraints including the possible linkages between tables using foreign key relationships.
- 2) Write relational algebra to find the students who worked on projects guided by faculty from CSE department.
- 3) Write relational algebra to find students who worked on all projects in 'AI' domain.

Q.4 (A) Consider the relational schema provided in the above section to answer the below (2)
questions using tuple relational calculus.

- 1) List all student names and their project titles (include students with no projects).
- 2) Find domains where no project is guided by any faculty.

Q.4 (B) Consider the relational schema provided in the above section to answer the below (3)
questions using SQL queries.

- 1) Print students name and ID who never worked on a project guided by 'ECE' faculty.
- 2) Print the count of projects per student and show only students with more than 2 projects.
- 3) Print the faculty name and ID who guide more than 3 projects in the same year.